

FULL NAME: _____

BLOCK: _____

Solving Radical Equations — Foundation

Lesson 5-4 · Period 1 · Student Packet

OBJECTIVES

Math Objective	Solve radical equations by isolating the radical and raising both sides to the appropriate power; identify and reject extraneous solutions.
Language Objective	Explain in writing, using sentence frames, why squaring both sides can introduce extraneous solutions and how to check.
Essential Understanding	Squaring both sides is NOT a reversible operation — it can create solutions that don't satisfy the original equation. Every candidate must be checked.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Solve single-radical equations; detect extraneous solutions.
Essential Question	Why does squaring both sides sometimes produce solutions that don't work, and how do we catch them?
Materials	Student packet, pencil. Teacher models Example 1, then gradual-release into Example 3 (extraneous).

LAUNCH — Solving Radical Equations (teacher models on screen)

[BOOK] ISOLATE–RAISE–CHECK RULES (keep printed on your page)

1. Isolate the radical on one side of the equation.
2. Raise BOTH sides to the index power (square for $\sqrt{}$, cube for $\sqrt[3]{}$, etc.).
3. Solve the resulting polynomial equation.
4. CHECK each candidate in the ORIGINAL equation — reject any that don't satisfy it (extraneous).

[WARN] COMMON ERROR

Do NOT skip the CHECK step — squaring both sides is not a reversible operation. A candidate solution that makes the radicand negative under an even-index radical is extraneous. Always substitute back into the ORIGINAL equation, not the squared form.

EXPLORE — Think-Pair-Share (33 min)

Work with a partner. Use the Isolate–Raise–Check Rules above for every problem. Move through items in order — each builds on the last.

Bank-item labels (in order):

1. Try It 1 — Isolate and raise to solve

2. Try It 3 — Solve and check for extraneous solutions
3. Practice #22 — Solve radical equation
4. Practice #8 — Solve radical equation
5. Practice #15 — Identify extraneous solutions
6. Practice #4 — Solve and verify

Try It 1 — Isolate and raise to solve

Solve each radical equation. Check your solution by substituting into the given equation or graphing.

- a. $\sqrt{x-2} + 3 = 5$
- b. $\sqrt[3]{x-1} = 2$

Try It 3 — Solve and check for extraneous solutions

Solve each radical equation. Check for any extraneous solutions.

- a. $x = \sqrt{7x+8}$
- b. $x+2 = \sqrt{x+2}$

Practice #22 — Solve radical equation

Solve each radical equation. ($\sqrt{4x} = 11$)

Practice #8 — Solve radical equation

$$\sqrt{8x+9} = x$$

Practice #15 — Identify extraneous solutions

Generalize Explain how to identify an extraneous solution for an equation containing a radical expression.

Practice #4 — Solve and verify

Error Analysis Nazim said that -3 and 6 are the solutions to $\sqrt{3x+18} = x$. What error did Nazim make?

SHARE / SUMMARY (5 min)

Sentence frames

Pair share: "I isolated the radical to get _____. Squaring both sides gives _____. My candidate solutions are _____. When I check in the ORIGINAL equation, _____ is valid and _____ is extraneous because _____."

Whole class: one pair reads their sentence. Class signals thumbs up/sideways.

EXIT

To solve a radical equation I first ____.

I raise both sides to power ____ because ____.

An extraneous solution happens when ____.

One biggest thing I learned today: _____

OPTIONAL REINFORCEMENT (not collected)

If you finish Explore early or want extra practice before Period 2.

[SEARCH] OPTIONAL • Model & Discuss

Return to the Do Now equation. With your partner:

B) Use Structure. Why can squaring both sides introduce a solution that doesn't work in the original equation?

C) Examine at least two equations you solved in Explore. For each, explain in your own words why checking is a required step and not optional.

Optional #1 (Savvas Practice)

Solve each radical equation. ($\sqrt[3]{x} + 8 = 13$)